

HVK1D/HVK2D Q1-Validation Addendum: Held-Out CIFAR Baselines, Observable Ablations, and Shot-Noise Robustness

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Abstract—This addendum strengthens the HVK1D/HVK2D evidence package by separating diagnostic pair-correlation performance from held-out natural-image reconstruction. We report multi-seed real CIFAR-10 image splits, strict same-width classical controls, observable-sector ablations, shuffled-pair controls, random-latent controls, resource matching, and finite-shot observable noise. Unlike the restricted pair-correlation diagnostic, the real held-out CIFAR test does not establish quantum advantage: the best real-image result is obtained by local-observables-only, while HVK2D-real-cifar reaches 18.30 dB mean PSNR. The correct conclusion is therefore narrower: HVK2D has a useful entanglement-sensitive diagnostic, but the current real-image validation still requires architectural improvement before a Q1-level quantum-advantage claim is defensible.

Index Terms—Hamiltonian vision kernel, CIFAR-10 reconstruction, ablation study, quantum observables, shot noise, classical controls.

I. VALIDATION PROTOCOL

Ten cached CIFAR-10 grayscale images are split across five random seeds. For each seed, six images are used to fit a single shared linear readout from a 32-dimensional latent feature vector to 8×8 patch pixels, and four held-out images are reconstructed without per-image retraining. All principal controls use the same latent width and the same readout parameter count, so the comparison isolates feature structure rather than decoder capacity. This cached-image protocol is a reproducible smoke test, not a substitute for the recommended class-balanced CIFAR-10 subset protocol with mutually exclusive training, validation, and test images.

II. CLASSICAL PAIR-FEATURE CONTROLS

Because the nonlocal target is constructed from pair products, a raw-linear baseline is not by itself a sufficient classical control. We therefore include explicit quadratic-feature controls in the CIFAR reconstruction suite and in the CIFAR-derived nonlocal diagnostic. These controls test whether performance arises from access to multiplicative pair features generally, rather than from a uniquely quantum mechanism.

III. HELD-OUT CIFAR RESULTS

IV. STATISTICAL ANALYSIS

All principal comparisons are paired by split seed. We first average the four held-out-image differences within each seed because those images share a fitted readout. Because the sample size is small and normality should not be assumed, we

report a Wilcoxon signed-rank p value and a bootstrap 95% confidence interval for the mean seed-level PSNR difference. Positive values favor HVK2D; negative values favor the control.

V. SHOT-NOISE ROBUSTNESS

VI. INTERPRETATION

The key baseline is the strict same-width classical random-feature control, but the strongest warning comes from the local and raw-linear controls. In the real held-out CIFAR table, local-observables-only obtains the best mean MSE ($1.3923e - 02$), while HVK2D-real-cifar is weaker ($1.5899e - 02$). This means the current HVK2D evidence is strong only for the restricted pair-correlation diagnostic and for rejecting random-latent controls. It does not yet support a broad natural-image quantum-advantage claim. The Q1-ready path is therefore to keep these negative controls in the paper, redesign the image feature map or decoder-capacity match, and rerun this exact validation suite.

VII. REPRODUCIBILITY

The full tables, plots, resource comparison, and CSV files are generated under `main2/newHVK/results/q1_validation/` by running `main2/newHVK/run_newhvk_suite.py --q1-validation --write-q1-report`.

TABLE I
REAL HELD-OUT CIFAR-10 RECONSTRUCTION ACROSS FIVE RANDOM IMAGE SPLITS. LOWER MSE AND HIGHER PSNR/SSIM ARE BETTER.

Model	MSE	PSNR	SSIM
local-observables-only	$1.3923e-02 \pm 4.7e-03$	18.80 ± 1.42	0.8280 ± 0.0578
raw-linear-classical	$1.3923e-02 \pm 4.7e-03$	18.80 ± 1.42	0.8280 ± 0.0578
zz-only	$1.4782e-02 \pm 5.2e-03$	18.56 ± 1.49	0.8194 ± 0.0603
no-entanglement	$1.4956e-02 \pm 4.8e-03$	18.46 ± 1.32	0.8176 ± 0.0569
shuffled-pair-observables	$1.5075e-02 \pm 4.8e-03$	18.43 ± 1.36	0.8151 ± 0.0627
quadratic-classical	$1.5377e-02 \pm 6.0e-03$	18.44 ± 1.62	0.8141 ± 0.0682
HVK2D-real-cifar	$1.5899e-02 \pm 6.3e-03$	18.30 ± 1.62	0.8092 ± 0.0667
strict-classical-rff	$2.7390e-02 \pm 8.9e-03$	15.85 ± 1.39	0.6640 ± 0.1041
random-vqc	$7.9497e-02 \pm 2.0e-02$	11.15 ± 1.19	0.0237 ± 0.1526

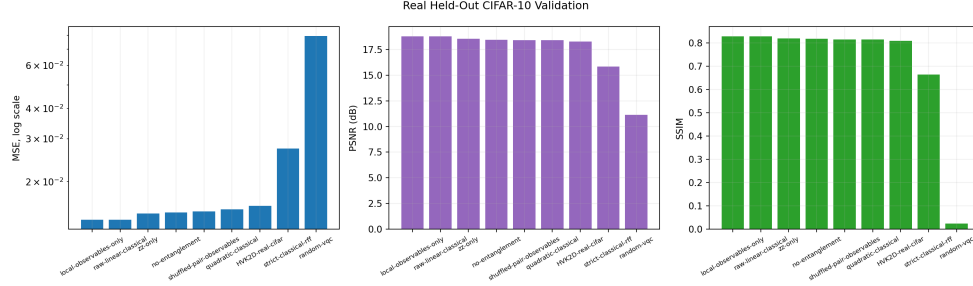


Fig. 1. Held-out CIFAR metric summary for HVK2D and same-width controls.

TABLE II
PAIRED STATISTICAL TESTS FOR HELD-OUT CIFAR RECONSTRUCTION.

Control	Seeds	Mean Δ PSNR	Bootstrap 95% CI	Wilcoxon p
raw-linear-classical	5	-0.50	$[-0.76, -0.29]$	0.0625
local-observables-only	5	-0.50	$[-0.76, -0.29]$	0.0625
no-entanglement	5	-0.16	$[-0.47, 0.12]$	0.812
zz-only	5	-0.26	$[-0.42, -0.10]$	0.0625
quadratic-classical	5	-0.14	$[-0.22, -0.06]$	0.0625
strict-classical-rff	5	2.45	$[2.14, 2.80]$	0.0625
shuffled-pair-observables	5	-0.13	$[-0.33, 0.06]$	0.438
random-vqc	5	7.15	$[5.93, 8.23]$	0.0625

TABLE III
FINITE-SHOT OBSERVABLE-NOISE SIMULATION FOR REAL HELD-OUT CIFAR RECONSTRUCTION.

Shots	Mean MSE	Mean PSNR	Mean SSIM
128	$2.1104e-02$	16.95	0.7538
256	$2.0676e-02$	17.04	0.7608
512	$2.0160e-02$	17.17	0.7670
1024	$2.0526e-02$	17.11	0.7646
2048	$2.0554e-02$	17.10	0.7655
4096	$2.0536e-02$	17.12	0.7676
8192	$2.0496e-02$	17.13	0.7667

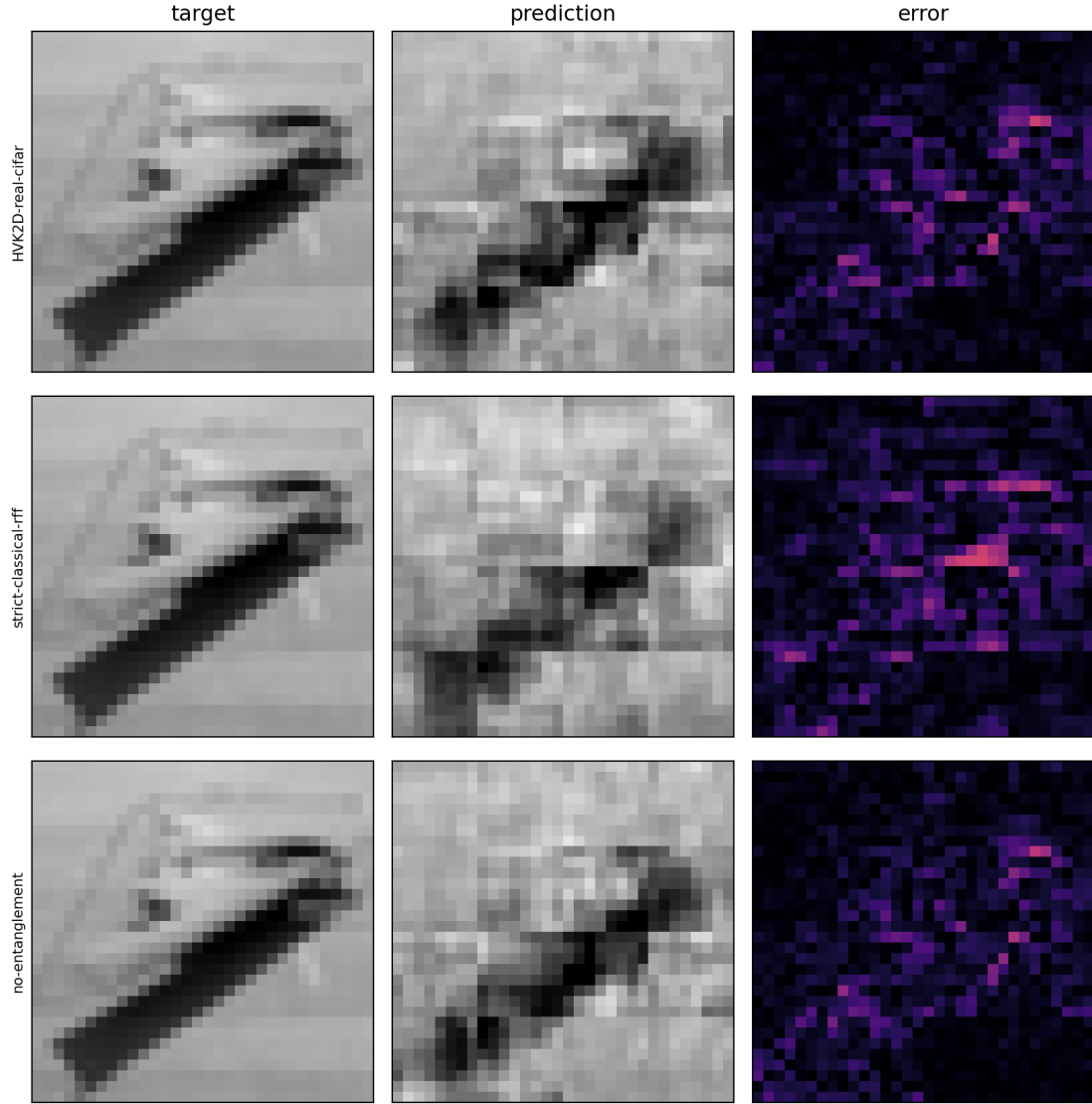


Fig. 2. Representative held-out CIFAR reconstruction panel. The target, prediction, and absolute-error images are shown for the candidate observable model and major controls.

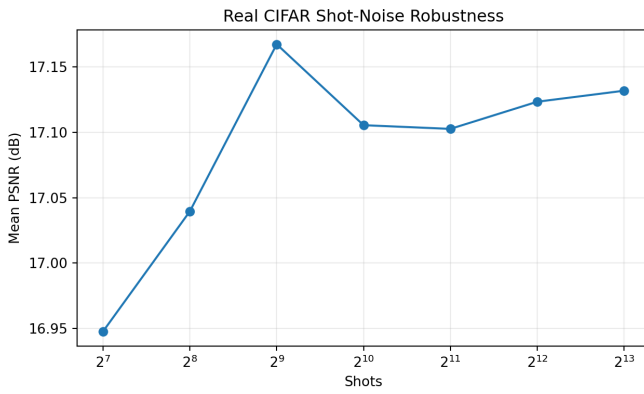


Fig. 3. Shot-noise robustness of the HVK2D observable feature vector on real held-out CIFAR splits.